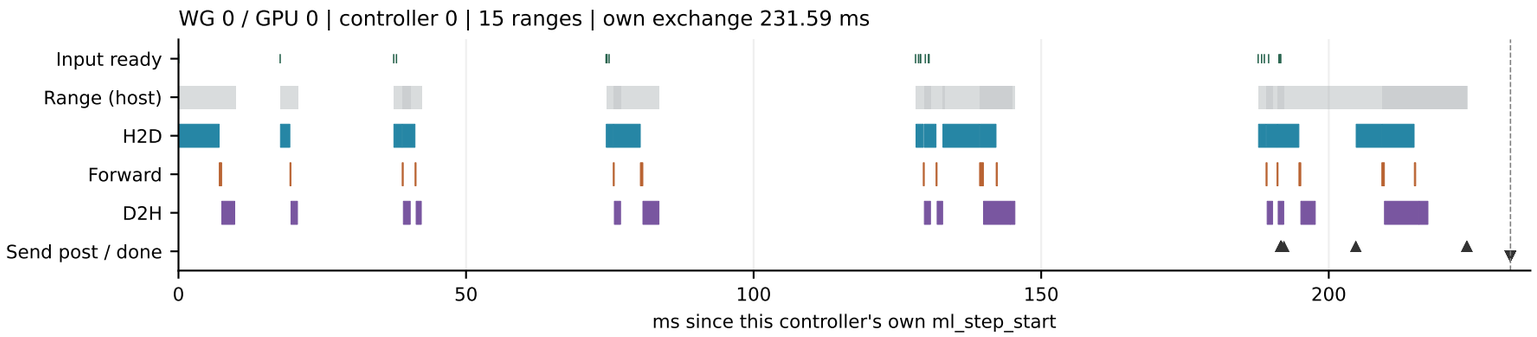
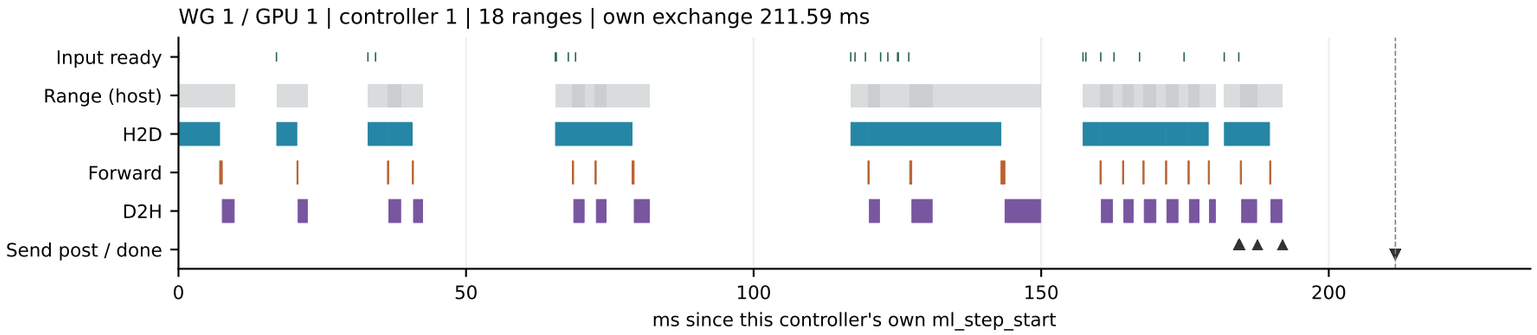
mlp | 96c4g | default | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 381.51 ms | p2p1 395.21 ms



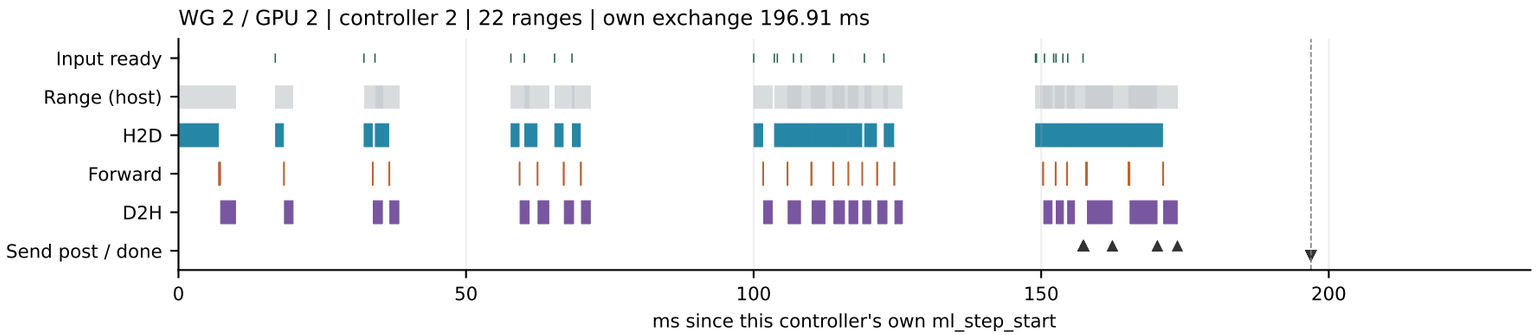
GPU work (ms)	Collective	p2p1
H2D	29.368	47.531
Forward	2.891	4.231
D2H	15.471	28.873
GPU SUM	47.730	80.635

Controllers: coll 0 / P2P 0
Coll host device elapsed: 47.878 ms
P2P GPU envelope: 217.115 ms
Different scopes; not an elapsed-time comparison.



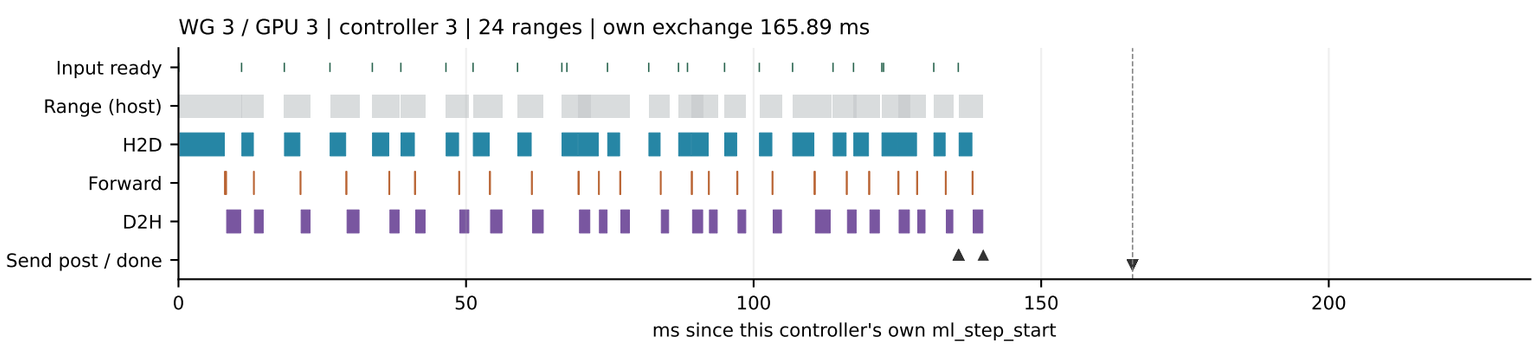
GPU work (ms)	Collective	p2p1
H2D	53.826	86.556
Forward	2.901	4.782
D2H	25.652	39.666
GPU SUM	82.379	131.004

Controllers: coll 1 / P2P 1
Coll host device elapsed: 82.519 ms
P2P GPU envelope: 191.754 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	22.888	60.482
Forward	2.696	4.166
D2H	25.919	42.815
GPU SUM	51.503	107.463

Controllers: coll 2 / P2P 2
Coll host device elapsed: 51.730 ms
P2P GPU envelope: 173.547 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	27.359	65.867
Forward	2.699	4.644
D2H	25.424	39.850
GPU SUM	55.483	110.360

Controllers: coll 3 / P2P 3
Coll host device elapsed: 55.653 ms
P2P GPU envelope: 139.616 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.